CA6001 – AI Algorithms Fundamentals & Application

Group Project Technical Report

Machine Learning–Guided Adaptive CPT Site Investigation and Economic Benefit Assessment

Predicting shallow cone-resistance profiles, choosing the next tests, deciding when to stop, and pricing the outcome

Project Title: Machine Learning–Guided Adaptive CPT Site Investigation and Economic Benefit Assessment

Course: CA6001 AI Algorithms Fundamentals & Application

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Student Names and Contributions:

Name	Matriculation no.	Main contributions

Code, configuration and result files: project repository, package `cpt_budget`; figures and tables in this report are regenerated from the saved results.

Summary. Cone penetration tests (CPTs) measure ground resistance along single vertical lines, so the ground between tests must be inferred. We ask whether AI models can predict unmeasured CPT profiles from a few measured ones, choose where to test next and decide when to stop, and what that is worth. Five predictors – a Gaussian process (GP), XGBoost, a conditional neural process (CNP) and two TabPFN hybrids – are replayed on eight sites. Stopping thresholds were frozen on four development sites, and every stop was audited against hidden CPTs. The TabPFN hybrids gave the lowest profile error, but within about 3% of XGBoost and GP. On the four evaluation sites no early stop failed the audit (7 early stops in 24 runs). However, CPTs were saved only at the most uniform site (21 → 11, the minimum the rule allows). Adaptive location choice showed no detectable advantage over a space-filling baseline. In the economic scenario, time effects outweighed CPT prices: the net owner benefit was +€23,300 at that site and negative elsewhere.

1. Introduction

Domain. Cone penetration tests (CPTs) push an instrumented cone into the ground and record the cone resistance q_c (in MPa) with depth: higher q_c means stiffer ground, and low-q_c layers indicate soft material that matters for design. CPTs are the main source of shallow ground information on civil projects, but each test samples only one vertical line, so the ground between tests is inferred. A conventional investigation fixes the number and positions of CPTs in advance and completes them all; some add little new information, while extra tests are usually ordered by judgement.

Motivation and objective. We ask: *with only a few CPTs measured, can an AI model predict the unmeasured profiles well enough to choose the next most useful CPTs and to stop once quality is adequate – and what is that worth?* Fewer CPTs reduce direct cost, and finishing earlier can bring a whole programme forward. The objectives are to (i) compare five AI predictors under identical information; (ii) wrap the selected predictor in an adaptive selection and online stopping loop; (iii) audit every stop against hidden CPTs that the loop never sees; and (iv) translate the resulting CPT counts into an owner's financial scenario.

Contribution. The project provides a replay framework that evaluates prediction, location choice, stopping and economic value together under one information boundary (Fig. 1). It reports negative results as well as savings.

Scope. We use public CPT data in the 2–8 m depth window. The threshold $q_c < 1$ MPa is a research indicator of soft ground, not a soil-classification, bearing-capacity or liquefaction criterion. The economics are counterfactual scenarios, not observed contract savings.

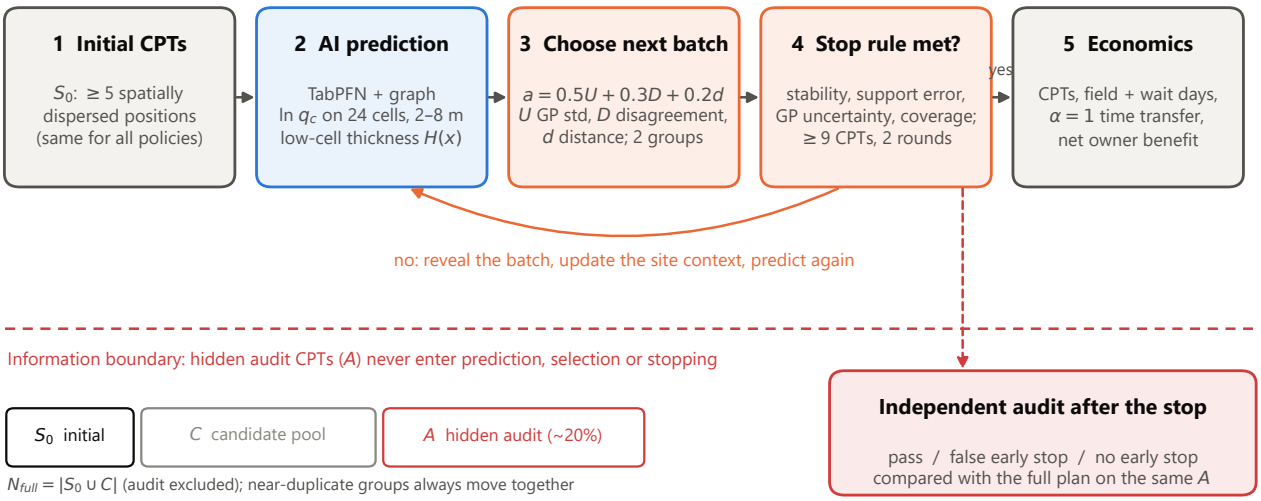


Figure 1: Investigation loop. Revealed CPTs feed prediction, batch selection and an online stop rule. The hidden audit set A is read only by a separate evaluator after the stop. All policies share the initial set S_0, the candidate pool C and A.

2. Problem Definition

At a site with planned CPT positions x and 24 depth cells $z_k = 2.125, 2.375, \ldots, 7.875$ m (each 0.25 m thick), the regression target is $y(x, z_k) = \ln(q_c/1\,\text{MPa})$. A cell is a *low cell* when $q_c < 1$ MPa, i.e. $y < 0$, and the cumulative low-cell thickness at a position is

$$H(x) = 0.25 \sum_{k=1}^{24} \mathbb{1}[y(x, z_k) < 0] \quad \text{(m)}. \tag{1}$$

Given the revealed CPTs, a model predicts $\hat{y}(x, z)$ and $\hat{H}(x)$ at every unrevealed position. Each site is split into an initial set S_0, a candidate pool C and a hidden audit set A. Starting from the revealed set $R = S_0$, a policy repeatedly picks a batch from $C \setminus R$, reveals it, and decides whether to stop. It seeks the smallest stopping count N_{stop} for which quality on A meets frozen targets, compared with completing the full plan of $N_{\text{full}} = |S_0 \cup C|$ CPTs:

$$\min\ N_{\text{stop}} \quad \text{s.t.} \quad \text{log-MAE}_A \le \tau_{\log},\ \ \text{thickness-MAE}_A \le \tau_H,\ \ \text{loss relative to the full plan} \le \varepsilon. \tag{2}$$

The task therefore combines several problem types: (i) *spatial regression*: predicting the profile; (ii) *rare-event classification*: detecting low cells, evaluated with precision, recall, F1 and false-positive rate (FPR, the share of truly non-low cells predicted low); (iii) *active learning*: choosing informative locations; (iv) *optimal stopping*: deciding when to stop.

Constraints:

- **Information:** only coordinates, planned depths and already revealed curves may enter prediction, selection or stopping. Hidden labels are read by a separate evaluator after the stop.
- **Grouping:** near-duplicate or repeated soundings form indivisible groups. A nominal batch of two can therefore be larger, and all counts are actual positions.
- **Small data:** sites, not depth cells or random seeds, are the replicate unit. With four evaluation sites the results are descriptive, and no significance tests are claimed.
- **No hindsight:** all thresholds are fixed on development sites before evaluation. An online stop that fails the audit remains a failure.

3. Dataset Description

Source. The data come from the public CPT databases of ISSMGE Technical Committee 304 (ISSMGE Technical Committee 304, n.d.). The Global-CPT collection holds 1,196 soundings from 59 sites or regions (Ching et al., 2023). The NL-CPTU/98/2km^2 set contains the 98 soundings of the New Lock Terneuzen ship lock in the Netherlands, fully operational since 1 August 2025 (Rijkswaterstaat, 2025). Only the subset that supports spatial prediction is used (Table 1). Of 31 sites, 23 are *source* sites that only train the models. The 8 *replay* sites, those with at least 20 usable positions, are split into 4 development and 4 evaluation sites for the decision loop.

Table 1: Data hierarchy after screening. Depth cells are repeated measurements within a CPT, not independent samples.

Unit	Count	Meaning
Raw profiles	503	CPT curves with coordinates that cover 2–8 m
Distinct planned positions	495	repeated soundings at one coordinate merged (8 removed)
Usable positions	492	complete 24-cell window (3 incomplete profiles removed)
Near-duplicate groups	456	435 singletons; groups of 2–5 always move together
Sites	31	23 source-only + 8 replay sites (4 development, 4 evaluation)
Depth-cell records	11,808	492×24 cells of 0.25 m

Features and preprocessing. Each record carries the site, local coordinates (x, y) in metres, near-duplicate group, depth and q_c. Each raw curve is interpolated to the 24 cell mid-points, q_c is log-transformed, non-representative repeats are removed and groups are assigned. For New Lock, 80 of the 98 original files are retained: 15 start below 2 m and 3 end above 8 m. The neighbour features used by the models are rebuilt from revealed CPTs only, at every round. They are distances, local medians, low quantiles, low-cell fractions and vertical gradients.

Exploratory analysis. The 0.25 m cells lose some thin layers. Against exact threshold crossings of the raw curves, the cell representation has a thickness MAE of 0.092 m (90th percentile, P90, 0.256 m). 498 of 805 raw low intervals thinner than 0.25 m contain no cell mid-point. All models nevertheless use the same grid, so comparisons stay fair.

The sites differ sharply (Fig. 2). Wellington Airport is uniformly stiff with no low cells, while New Lock has 24% low cells and strong lateral variability. Low cells make up only about 8% of source-site cells. Three properties follow: (i) *small samples per site*, which favour methods that borrow strength across sites or condition on context; (ii) *spatial correlation*, which favours kernel or neighbour-based models; (iii) *rare events*, which favour an explicit low-cell decision rather than a median predictor. These three properties drove the choice of algorithms in Section 4.

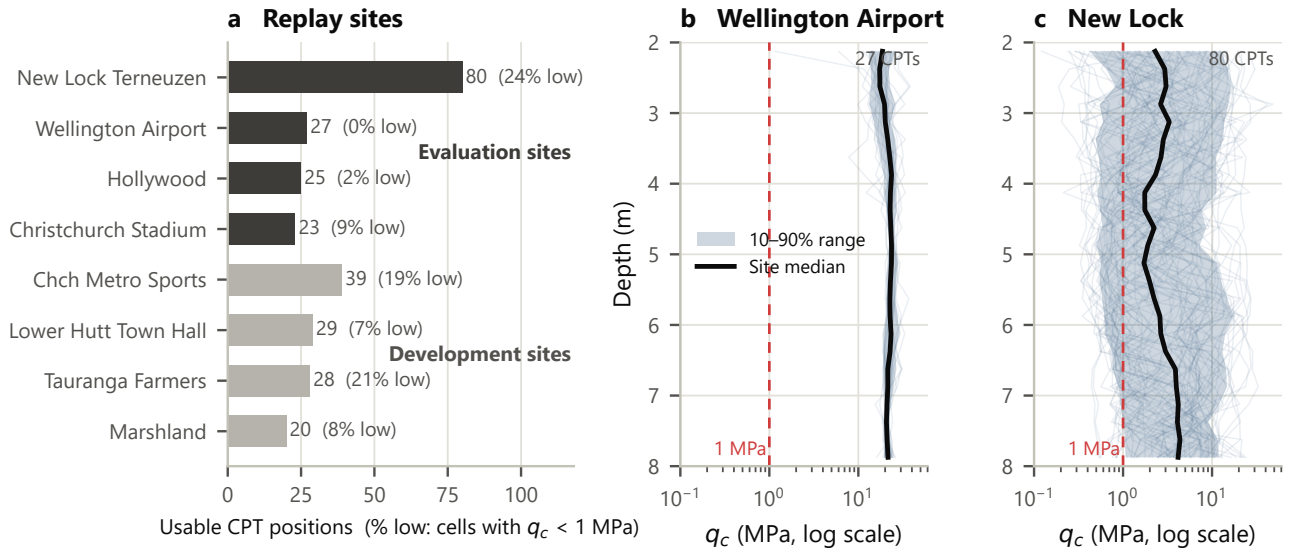


Figure 2: Replay sites and profiles. (a) Usable CPT positions at the eight replay sites and their share of low cells (q_c < 1 MPa). (b, c) All 2–8 m profiles at the most uniform (Wellington Airport) and most variable (New Lock) evaluation sites. The band is the 10–90% range and the black line the site median; the x-axis is logarithmic.

4. Algorithm Selection and Rationale

The five predictors span the main families for small-data spatial prediction. All of them receive the same revealed CPTs at every state. Thickness and low-cell labels are always derived from each method's single predicted curve.

Gaussian process (GP). A GP places a prior over functions whose covariance is set by a kernel. After conditioning on the revealed data, the predictive mean is a kernel-weighted combination of the observations, and the predictive variance grows with distance from them (Rasmussen & Williams, 2006). We use an anisotropic Matérn-3/2 kernel on (x, y, z) plus white noise. It models residuals from a source-site depth trend and is refitted each round. *Strength:* a native uncertainty for acquisition. *Limitation:* stationarity, which smooths thin low layers.

XGBoost residual. Gradient boosting adds shallow regression trees one at a time, each fitted to the current residuals (Chen & Guestrin, 2016). Here the trees correct a 3-nearest-neighbour inverse-distance (IDW) estimate using 24 neighbour features. *Strength:* non-linear local corrections. *Limitation:* the trees are fixed after source training.

Conditional neural process (CNP). A CNP encodes each revealed profile together with its position relative to the query. It pools the encodings with a permutation-invariant mean and max, and decodes a full profile at the query (Garnelo et al., 2018). It learns site-to-site transfer from many training "episodes". *Limitation:* its L1 loss targets the conditional median, so rare low cells are smoothed away.

TabPFN. TabPFN is a Transformer pre-trained on millions of synthetic tabular tasks. It predicts by *in-context learning*: the current site's revealed rows are passed as context, and no gradient step is needed per round (Hollmann et al., 2025). We fine-tuned the v2 weights on source sites, then froze them. Two hybrids are tested: (i) *TabPFN joint* averages a sign-guarded TabPFN residual with a triangulation estimate, then sets the low-cell count by a whole-profile decision; (ii) *TabPFN + graph* averages the residual with a graph-based robust interpolation and labels a cell low if either branch does, which favours recall at the cost of false positives.

Compare and contrast. The four families trade off differently: (i) GP needs no source data and gives uncertainty, but assumes smooth, stationary ground; (ii) XGBoost and TabPFN exploit cross-site regularities and non-linearity, at the cost of source training; (iii) TabPFN adapts to a new site without retraining; (iv) CNP is the most data-hungry and the most sensitive to its loss function.

Alternatives considered. Ordinary kriging is close to the GP. Entropy-based Bayesian compressive sampling (Guan et al., 2022) targets liquefaction, not q_c profiles. Reinforcement learning or full Bayesian optimisation was rejected because four evaluation sites cannot support a learned policy without overfitting.

Division of labour. The primary predictor was chosen by a rule fixed before evaluation: the lowest development log-MAE. This selected TabPFN + graph (0.578), ahead of XGBoost (0.597) and TabPFN joint (0.598). It drives prediction and stopping. The GP supplies the uncertainty signal, and XGBoost is the disagreement partner. This is a division of labour, not a stacked ensemble.

5. Methodology

5.1 Evaluation design

The loop in Fig. 1 is replayed on historical data in a three-level, site-held-out design.

1. *Source training.* In each of three folds the models are trained on source sites. Checkpoints and early stopping are chosen on that fold's validation sites.
2. *Development.* Four replay sites select the primary model, the acquisition weights and the stop thresholds: Lower Hutt Town Hall, Tauranga Farmers, Christchurch Metro Sports and Marshland.
3. *Evaluation.* Four replay sites test the frozen configuration: Wellington Airport, Christchurch Stadium, Hollywood and New Lock. Each is predicted with the fold weights that excluded it from both training and checkpoint selection.

For every site and seed (17, 29, 43), whole groups are assigned to the audit set A (≈20%), a spatially dispersed S_0 (≥5 positions) and the candidate pool C. Three policies share each split:

- **Full plan** (B_{full}) completes every position in $S_0 \cup C$; this is the engineering reference.
- **Sequential** (B_{seq}) adds the two candidates that maximise the minimum distance to any revealed CPT. This *maximin* rule is a space-filling baseline. It applies the stop rule.
- **Adaptive** (P_{adapt}) adds the two candidates with the highest acquisition score and applies the same stop rule.

Full versus Adaptive measures the total benefit. Sequential versus Adaptive isolates what intelligent location choice adds beyond simply stopping early. The evaluation therefore comprises 4 sites × 3 seeds × 2 stopping policies = 24 trajectories, plus 12 full-plan runs.

Metrics. All metrics are computed on A. (i) *log-MAE* is the mean $|\hat{y} - y|$ over audit cells; (ii) *thickness MAE* is the mean $|\hat{H} - H|$ over audit positions; and (iii) the *low-cell metrics* are precision, recall, F1 and FPR. Values are averaged within seed, then within site, and then equally across sites (the *equal-site mean*).

5.2 Acquisition score

For each candidate group the policy computes three signals: (i) U, the mean GP predictive standard deviation of $\ln q_c$; (ii) D, the share of cells where the primary model and XGBoost disagree on $q_c < 1$ MPa; (iii) d, the distance to the nearest revealed CPT. Each signal is converted to a 0–1 rank within the pool, and the score is

$$a(x) = 0.5\,\text{rank}(U) + 0.3\,\text{rank}(D) + 0.2\,\text{rank}(d). \tag{3}$$

The weights are an engineering heuristic: uncertainty first, model disagreement second and spacing as a tie-breaker. The only alternative tried omitted disagreement (0.7, 0, 0.3). Within a batch only d is updated after each pick, so the second choice is pushed away from the first without seeing its curve.

5.3 Online stopping and independent audit

Each round, the controller computes four signals on a fixed grid of all planned positions × 24 depths:

- the 90th percentile over grid cells of the round-to-round change in predicted $\ln q_c$ (ΔQ_{90});
- the 90th percentile over positions of the change in predicted thickness (ΔH_{90});
- a leave-group-out log-MAE on up to five initial groups;
- the 90th percentile of the GP standard deviation.

The controller stops when all four pass for two consecutive rounds, provided at least nine CPTs are revealed and every populated quadrant of the site is covered. Because batches add two CPTs, the earliest possible stop is at 11 CPTs. Thresholds were set as quantiles (0.50/0.75/0.90) of the signals at acceptable development states. The 0.50 quantile was selected because it maximised verified saving minus false-stop rate; the 0.75 and 0.90 quantiles gave development false-stop rates of 17% and 58%.

The audit has two parts: (i) *absolute targets* of log-MAE ≤ 0.626 and thickness MAE ≤ 0.881 m, the development full-plan 75th percentiles plus fixed margins; and (ii) a *relative tolerance* of no more than 0.05 log-MAE or 0.25 m worse than the full plan. Low-cell recall ≥ 0.5 with FPR ≤ 0.1 is audited separately as a diagnostic.

5.4 Hyperparameters

Table 2 lists every setting. Model settings were fixed during source training on validation sites. Decision settings were fixed on development sites. No setting was tuned on evaluation sites.

5.5 Economic model

The owner's net benefit of a policy relative to the full plan is

$$B_{\text{net}} = (C_{\text{full}} - C_{\text{policy}}) + \alpha\,\Delta T_{\text{inv}}\,(c_{\text{day}} + b_{\text{day}}) - C_{\text{AI}}, \tag{4}$$

where the terms are: (i) C is investigation cost (€750 per CPT, all-in); (ii) ΔT_{inv} is the calendar days saved: 5 CPTs per rig-day with one rig, converted at 7/5, plus a 0.25-day decision delay at every batch transition; (iii) c_{day} = €5,000 is the avoidable project cost per day; (iv) b_{day} = €3,000 is the owner's benefit per day of earlier opening; (v) C_{AI} = €600 is deployment and review per run. In the main scenario $\alpha = 1$, meaning every investigation day saved shortens the project. $\alpha = 0.5$ and 0.75 are sensitivity cases, alongside one-at-a-time parameter ranges. A failed early stop would be repaired to the full plan, with remobilisation (€560) and two days' delay.

Table 2: Hyperparameters and how they were chosen.

Component	Setting (selection)
GP	Matérn $\nu = 3/2$, anisotropic length scales initialised at (20, 20, 1) m, white-noise term; refitted by marginal likelihood each round
XGBoost residual	3-NN IDW base (power 2) + $w\times$trees; 24 features; depth 6, learning rate 0.3, early stopping (36–37 trees); $w \in \{0, 0.25, 0.5, 1\}$ chosen per fold (0.25, 0.25, 0.5)
CNP	64 hidden units, mean + max pooling, L1 loss on $\ln q_c$; 26,817 parameters; 500–1,200 steps, checkpoint by validation log-MAE
TabPFN	v2 weights (package 9.2.0), 1 estimator; fine-tuned with CRPS (continuous ranked probability score) + MSE, 500–600 updates; joint blend 0.5/0.5
Graph branch	4 nearest neighbours, vertical weight 0.15, pseudo-Huber $\delta = 0.35$
Acquisition	weights (0.5, 0.3, 0.2); batch of 2 groups; alternative (0.7, 0, 0.3) rejected on development
Stop rule	≥9 CPTs, 2 rounds; $\Delta Q_{90} \le 0.082$, $\Delta H_{90} \le 0$ m, support log-MAE ≤ 0.584, GP-std P90 ≤ 0.461 (0.50 quantile of development signals)
Experiment	seeds 17, 29, 43; audit share ≈20%; $\lvert S_0 \rvert \ge 5$

Only the remobilisation cost and the consultant rate behind C_{AI} come from a 2025 Dutch quotation (MOS Grondmechanica B.V., 2025). All other values are explicit assumptions, following cost- and schedule-estimating guidance (U.S. Government Accountability Office, 2015, 2020). The same Dutch-based values are applied to every site; for non-Dutch sites this is a standardised transfer, not local evidence.

5.6 Tools and libraries

The implementation uses Python 3.11, NumPy 2.4, pandas 3.0 and SciPy 1.17, scikit-learn 1.9 for the GP (Pedregosa et al., 2011), XGBoost 3.2, PyTorch 2.11 for the CNP, TabPFN 9.2.0 (Prior Labs, n.d.) and matplotlib. Models ran on an NVIDIA RTX 4080 (16 GB).

6. Implementation

Structure. The code is a Python package, `cpt_budget`. Its module boundaries mirror the information boundary: (i) `adaptive_data` builds the splits and manifests; (ii) `adaptive_models` gives all five models one interface, `condition(revealed)` and `predict(query)`, and strips every query of measured columns; (iii) `adaptive_policy` holds the acquisition score and the stop rule; (iv) `adaptive_replay` runs the loop; (v) a separate evaluator reads A; (vi) `adaptive_economics` prices the outcomes. One command runs each stage: prepare, calibrate, freeze, evaluate, publish and economics. The frozen configuration stores SHA-256 fingerprints of the data, weights and settings.

Main algorithm. The core of batch selection is shown below:

```
group["rank_U"] = rank01(group.uncertainty)        # GP std, fixed within the batch
group["rank_D"] = rank01(group.disagreement)       # low-cell disagreement
while remaining and chosen_count < batch_size:
    dist  = [min_distance(g, observed_xy) for g in remaining]
    score = 0.5*group.rank_U + 0.3*group.rank_D + 0.2*rank01(dist)
    gid   = argmax(score)                          # ties: group id, then location id
    chosen += members(gid)                         # the whole near-duplicate group
    observed_xy = vstack([observed_xy, xy(gid)])   # coordinates only, never curves
```

Each round is logged as one trajectory row. The adaptive run at Wellington Airport (seed 17) stops in round 3, the second consecutive passing round:

```
round n_obs dQ90   dH90 support_logMAE gp_std_P90 coverage streak stop
  3    11   0.067  0.00     0.132        0.169      ok       2   online_stability
```

Key decisions. (i) TabPFN weights are frozen during replay and only the site context changes, so a round takes seconds rather than a retraining run; (ii) grouping is enforced in selection, counting and auditing; (iii) the fine-tuned checkpoints differ from initialisation (all 14 CNP and 129 TabPFN weight tensors changed), so weak CNP results are not an untrained-model artefact.

Correctness checks. Automated checks confirm four things: (i) hidden labels cannot influence predictions (24 paired initial contexts, maximum prediction difference 0); (ii) group counts and audit exclusion are exact; (iii) frozen inputs are unchanged; (iv) every economic component recomputes from its parts. The project's 95 unit tests pass.

7. Results and Evaluation

7.1 Prediction quality of the five methods

Main comparison. TabPFN joint has the lowest evaluation profile error (Table 3): 0.497, 2.9% below XGBoost. TabPFN + graph is 1.8% below XGBoost. The margins are small and site-dependent. Joint beats XGBoost in 9 of 12 site–seed cells, by 0.9–3.9% per site. At New Lock the GP is better than every TabPFN hybrid (0.976 against 1.034). On evaluation sites the GP has the lowest thickness error (0.400 m), but mainly because it rarely predicts low cells: its recall is 14%. On

Table 3: Prediction quality (equal-site means over registered observation states). †Primary model, selected on development log-MAE. Bold marks the best of the five main methods per column; CNP's 0% FPR is not bolded because it never predicts a low cell. Recall excludes Wellington Airport, which has no low cells.

	Development (4 sites)		Evaluation (4 sites)				
Method	log-MAE	Thick. (m)	log-MAE	Thick. (m)	Recall (%)	FPR (%)	F1
GP	0.658	0.614	0.509	**0.400**	14.0	**1.9**	0.11
XGBoost residual	0.597	0.571	0.512	0.485	31.2	6.9	0.30
CNP	0.833	0.774	0.657	0.430	0.0	0.0	0.00
TabPFN joint	0.598	**0.483**	**0.497**	0.491	33.3	7.8	0.30
TabPFN + graph†	**0.578**	0.586	0.503	0.595	**36.3**	9.6	**0.31**
TabPFN ablation (the fine-tuned joint row above completes the set)							
Residual, pretrained weights	0.627	0.642	0.506	0.494	23.4	5.9	0.23
Residual, fine-tuned weights	0.598	0.626	0.502	0.509	22.4	6.0	0.22
Residual, fine-tuned + sign guard	0.598	0.573	0.503	0.497	32.1	7.3	0.30
Joint, pretrained weights	0.604	0.485	0.495	0.480	34.9	7.5	0.32

development sites its thickness error is the second worst. The graph union buys the highest recall (36%), at the cost of the highest FPR (9.6%) and the largest thickness error. CNP never predicts $q_c < 1$ MPa, a genuine negative result for a median-trained set model on rare events.

Site difficulty. Site difficulty dominates model differences (Fig. 3): log-MAE is 0.11–0.24 at Wellington Airport and 0.98–1.14 at New Lock.

TabPFN ablation. Source-site fine-tuning lowered the raw residual's development log-MAE from 0.627 to 0.598. On evaluation sites, however, the pretrained and fine-tuned joint models differ by under 0.002. Most of the thickness gain comes from the spatial blend and the whole-profile decision (development 0.626 → 0.483 m), not from the Transformer weights. The improvement belongs to the hybrid, not to TabPFN alone.

Error analysis. In the full-plan predictions at New Lock, error peaks at 4–6 m, where 23% of cells are low. In that band the primary model recovers 53% of low cells but raises 85 false alarms. The GP flags only 1–3% of cells as low, against 9–23% true, so it under-predicts soft layers throughout. Thin layers below the 0.25 m grid add irreducible error.

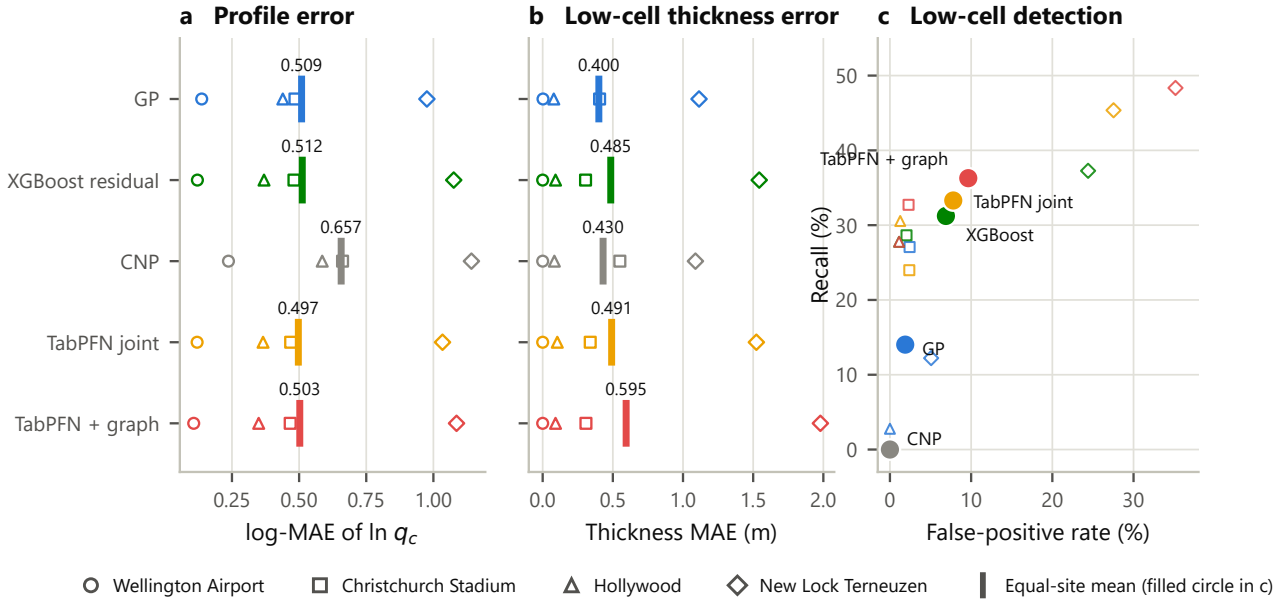


Figure 3: Five AI methods on identical observation states at the four evaluation sites. Open marks are site means (shape = site), and bars and filled circles are equal-site means. (c) Low-cell recall against FPR; recall is undefined at Wellington Airport.

7.2 Selection, stopping and audit

Where each policy samples. The adaptive score does choose different locations from maximin (Fig. 4). At New Lock it concentrates on the dense, variable central cluster where uncertainty and disagreement are high, whereas maximin spreads out.

Quality as CPTs are added (Fig. 5). At the three easier sites audit quality is inside the targets from the first few CPTs. At New Lock the error falls steeply over the first 15 CPTs and then plateaus at log-MAE ≈ 0.9 and thickness MAE ≈ 1.1–1.5 m, above both targets, even with all 64 CPTs.

Stopping outcomes. No early stop failed the audit (Table 4, Fig. 6), but the evidence is thin. Only 7 of the 24 evaluation

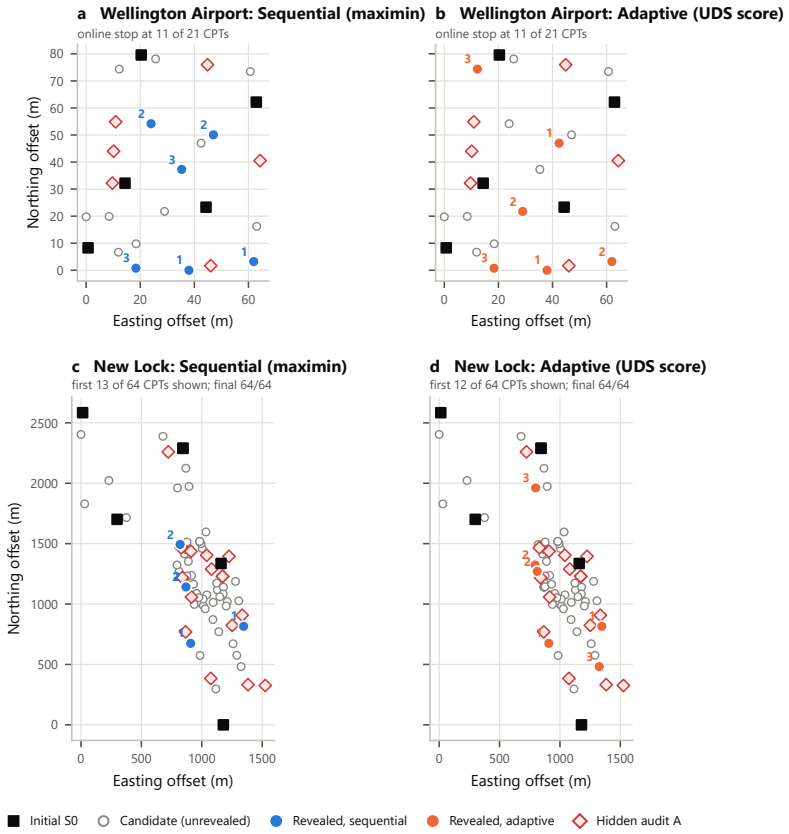


Figure 4: Selection maps, seed 17. (a, b) Wellington Airport at the online stop. (c, d) New Lock after the first 12–13 CPTs. Numbers give the round in which a position was revealed. The adaptive score combines GP uncertainty U, model disagreement D and distance d.

Table 4: Decision outcomes at the evaluation sites (means of three seeds). $|A|$ is the number of audit positions. "Full plan passes" counts the seeds in which even the full plan meets the audit targets. Oracle N is the smallest adaptive count that would have passed, found with hindsight; it is a diagnostic, not a policy. Net is the owner's benefit of the adaptive policy at $\alpha = 1$.

			CPTs at stop		Saving (%)		Full plan	Oracle	Net
Site	N_{full}	$\lvert A\rvert$	Seq.	Adapt.	Seq.	Adapt.	passes	N (adapt.)	k€
Wellington Airport	21	6	11.0	11.0	47.6	47.6	3/3	9	+23.3
Christchurch Stadium	18	5	18.0	18.0	0.0	0.0	3/3	9	−12.6
Hollywood	20	5	18.7	20.0	6.7	0.0	3/3	13–14	−11.9
New Lock	64	16	64.0	64.0	0.0	0.0	0/3	none	−55.9
Equal-site mean					13.6	11.9			−14.3

trajectories stopped early: 6 at Wellington Airport, where every audited state passes, and 1 at Hollywood (Sequential, seed 17). The rule's real value is that it never stopped at New Lock, where every early stop would have failed. A naive rule that stops at the ninth CPT would have made 7 false stops (all six New Lock runs plus one at Hollywood). Ablations show that the leave-group-out support error carries this safety: without it the evaluation runs make 3 (Sequential) and 2 (Adaptive) false stops, whereas dropping any other signal adds none.

Savings. Equal-site savings were 13.6% (Sequential) and 11.9% (Adaptive). Wellington Airport, 21 → 11 CPTs (47.6%), is the only adaptive saving, and Sequential saved exactly the same there. Eleven CPTs is the earliest stop the rule allows, so this saving reflects how uniform the site is, not fine-grained learning.

Adaptive against Sequential. Adaptive location choice showed no detectable advantage. Across all 24 site–seed pairs on the eight replay sites, the two policies' stopping counts differed only 3 times, and in both directions. At matched CPT counts the sign of the audit-error difference varies across sites and seeds.

Low cells and audit size. The low-cell targets (recall ≥ 0.5, FPR ≤ 0.1) failed at all three sites that contain low cells. Audit sets are small (5–6 positions, 16 at New Lock), which limits every site-level verdict.

The rule is safe but conservative. With hindsight, Christchurch Stadium already met the targets with 9 of 18 CPTs, and Hollywood with 13–14 of 20 (Adaptive). Yet both Adaptive runs went to the full plan. No single signal is to blame. At Christchurch Stadium, ΔQ_{90} and the GP uncertainty block the most rounds; at Hollywood, ΔQ_{90} and ΔH_{90} block about equally often. $\Delta H_{90} \leq 0$ allows changes at up to 10% of positions. Its value of zero is a quantile tie in the development data, not a deliberately chosen tolerance. GP uncertainty was only weakly related to actual audit error on evaluation sites (mean

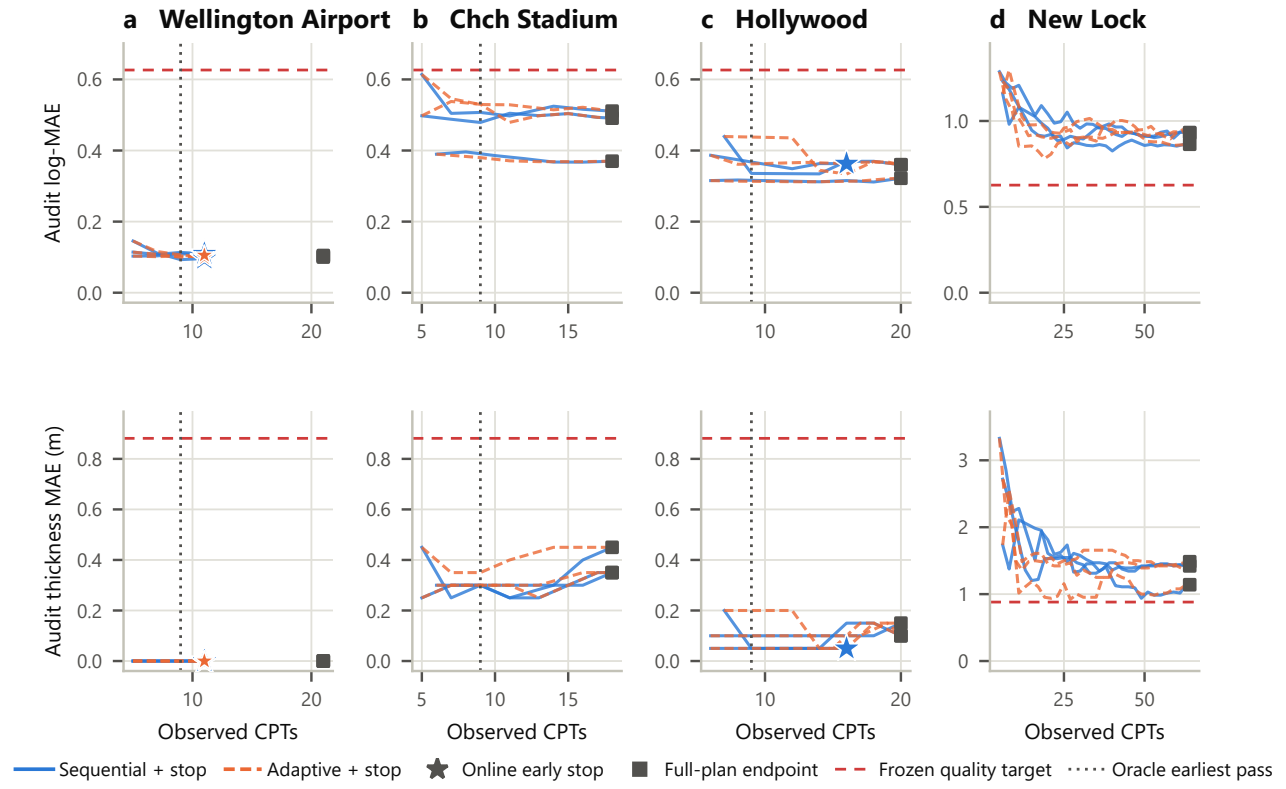


Figure 5: Hidden-audit quality as CPTs are added (three seeds per policy). Stars mark online early stops, squares the full plan, red dashed lines the frozen targets, and dotted lines the earliest passing count found with hindsight (none at New Lock). Step changes reflect the small audit sets.

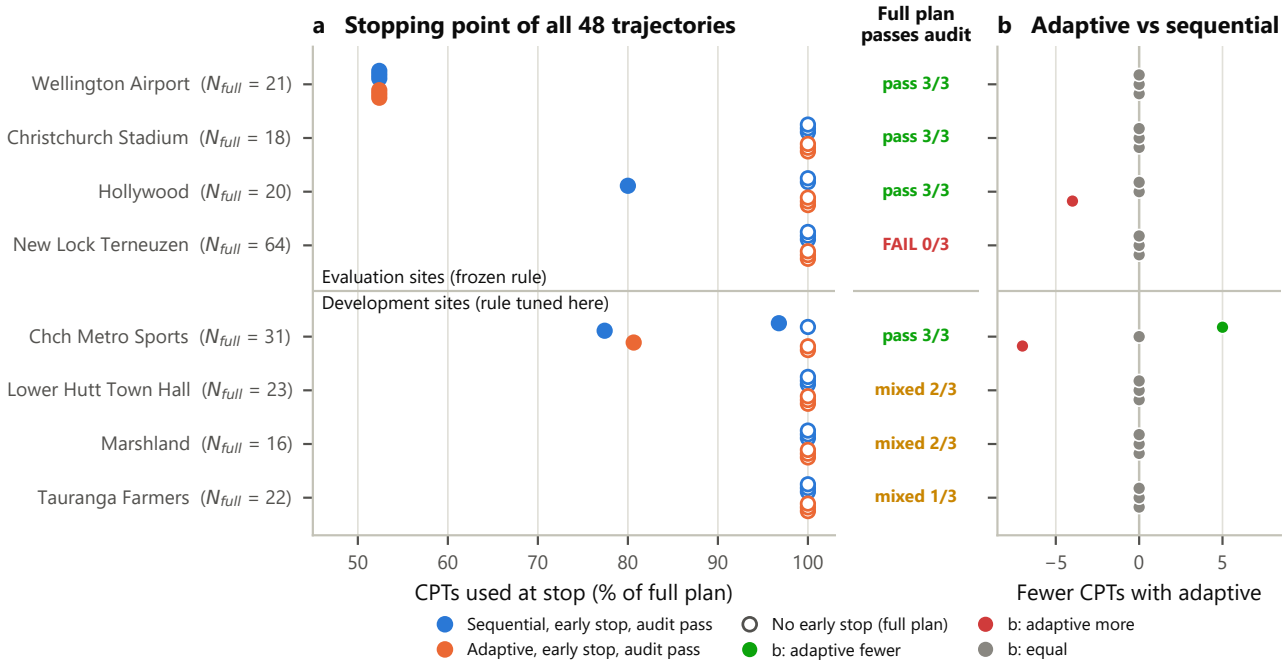


Figure 6: Stopping points of all 48 Sequential and Adaptive trajectories (8 sites × 3 seeds × 2 policies). (a) CPTs used at the stop, with the full-plan audit verdict per site. (b) CPTs saved by Adaptive relative to Sequential, paired by seed.

Spearman ρ = 0.21, against 0.60 on development sites).

7.3 Economic value

Wellington Airport. The observed counts were priced in the scenario (Fig. 7). The adaptive policy yields +€23,300: (i) €7,500 from ten avoided CPTs; (ii) €10,250 of avoided project cost and €6,150 of earlier-opening benefit, from finishing 2.05 days earlier; (iii) less €600 of AI cost. Sequential stopping earns the same amount, so this benefit comes from staged stopping, not from AI location choice.

Other sites. The other three sites are negative, from −€11,900 to −€55,900, although they used no extra CPTs. The staged campaign adds a 0.25-day decision delay at every batch transition. At New Lock there are 27–29 transitions, which delay

completion by 6.9 days. With zero delay these losses shrink to the €600 AI cost. With a one-day delay New Lock would lose €222,000, and Wellington Airport would still gain €5,300. Across all four evaluation sites the mean is −€14,300 for Adaptive and −€13,500 for Sequential.

Sensitivity. At $\alpha = 0.5$ Wellington Airport falls to €15,100. No site changes sign within any tested range.

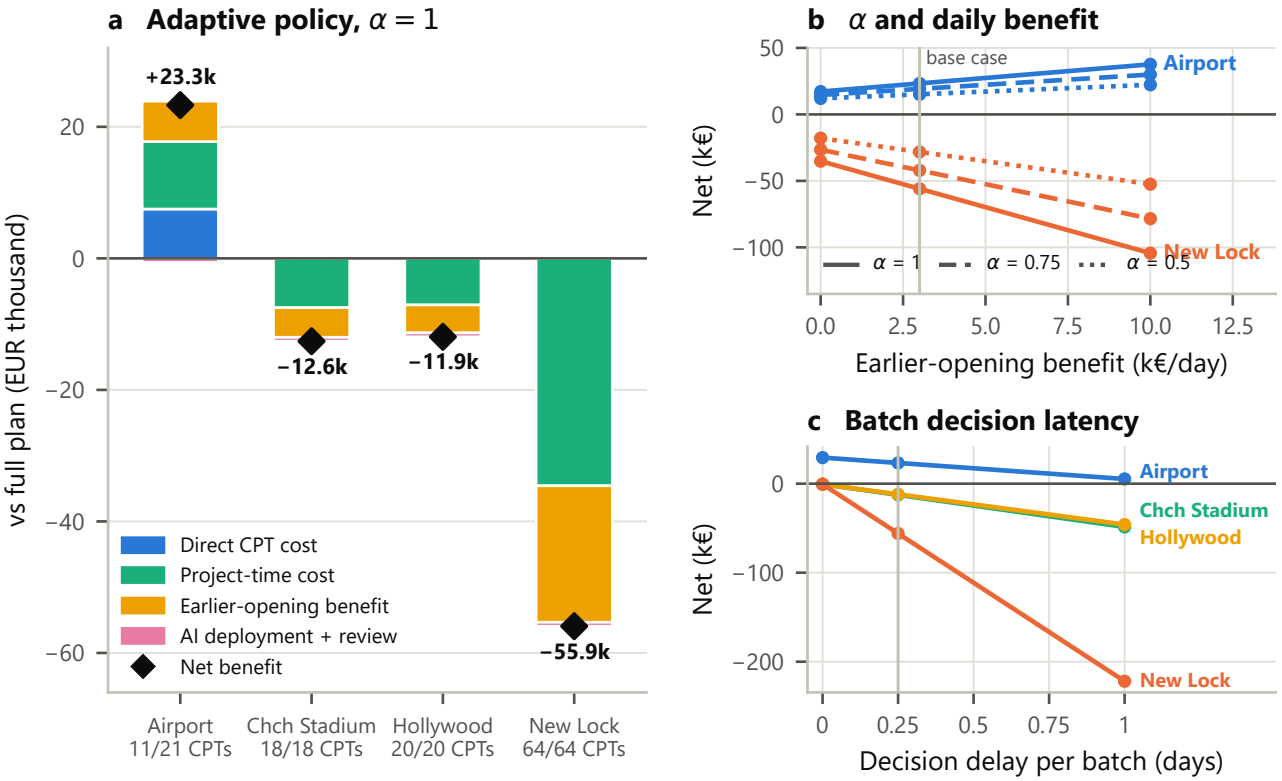


Figure 7: Owner's financial scenario, EUR 2025, Adaptive policy (mean of three seeds). (a) Benefit components relative to the full plan at $\alpha = 1$. (b) Net benefit as α and the daily earlier-opening benefit vary (Wellington Airport and New Lock; the vertical line marks the base case). (c) Net benefit against the decision delay per batch.

8. Discussion and Insights

What worked. The pipeline is auditable. Hidden labels never reach the controller, every stop is audited, and all trajectories are reported, including failures. No leakage was found by the automated checks. The stop rule vetoed every stop at the one site where stopping would have failed. The TabPFN hybrids were competitive, giving the best profile error and low-cell recall. The GP remained the most conservative low-cell detector and the source of the uncertainty signal.

What the results reveal. Four lessons emerge.

1. *Site predictability, more than the algorithm, decided whether CPTs could be saved.* Savings appeared only where five CPTs already describe the ground, and vanished at New Lock, where even 64 CPTs leave log-MAE near 0.9.
2. *Location choice added nothing measurable here.* The benefit came from staged investigation combined with a stopping rule.
3. *Model choice matters less than the decision it serves.* The five predictors lie within about 3% of each other on profile error. Their low-cell behaviour, however, differs sharply: GP is conservative, the graph hybrid sensitive and CNP blind. This difference changes what a stop rule can safely conclude.
4. *Time, not CPT price, dominates the economics.* Per-batch decision delay can turn a neutral outcome strongly negative. A deployed system must return decisions within the working day, and should recommend the full plan early when a site is clearly unpredictable.

Threats to validity.

- *Optimistic development data.* The four development sites served as checkpoint-selection (validation) sites during source training. This likely explains why the learned models beat the GP by 12% on development sites but by only 1% on evaluation sites. It also means the model ranking (0.578 vs 0.597) is within noise.
- *Prior exposure.* The evaluation sites had been inspected in an earlier project stage, so this is a site-held-out re-evaluation, not a blind test.
- *Small samples.* With four evaluation sites, 5–16 audit positions per site and seeds that share the same ground, all conclusions are descriptive. The 7 observed early stops give a 95% upper bound of roughly 40% on the false-stop rate.
- *Confounding at New Lock.* New Lock is the only non-New-Zealand site, the largest site, and has no close source analogue.

Domain shift, size and heterogeneity cannot be separated there.

- *Economic assumptions.* The economics rest on assumed unit values and on $\alpha = 1$, which places investigation on the critical path.
- *Representation and labels.* The 0.25 m grid misses thin layers, and the 1 MPa threshold is a research indicator.

Future improvements. (i) learn quality-predictive stop signals, rather than fixed quantile thresholds, so that the clear opportunities at Christchurch Stadium and Hollywood are taken; (ii) add a site-predictability check after the initial CPTs that recommends the full plan at once for New Lock-like sites; (iii) use larger batches or same-day decisions to cut latency; (iv) use an event-weighted loss and a 0.1 m grid for low-cell detection; (v) evaluate on genuinely new sites with real contract and programme data. A post-evaluation exploration (not used for any claim above) tried variance-reduction acquisition at New Lock. It lowered the GP's budget-averaged profile error by about 11% relative to maximin but worsened thickness error, and none of 432 explored states met the targets.

9. Conclusion

What was built. We built and audited an end-to-end, AI-guided CPT investigation loop. It combines five AI predictors, an uncertainty–disagreement–distance acquisition score, a multi-signal stop rule and an owner's economic model.

Findings. (i) the TabPFN hybrids gave the lowest profile error (0.497 log-MAE) and the highest low-cell recall (36%), within about 3% of XGBoost and GP; (ii) with rules frozen on development sites, no early stop failed the audit; (iii) CPTs were saved only at the most uniform site: 47.6%, at the minimum the rule allows, worth +€23,300 in the main scenario; (iv) adaptive location choice did not beat simple maximin ordering; (v) New Lock could not reach the quality targets with any number of CPTs.

Lessons. An AI algorithm should be judged by the decision it supports, using an independent audit and reporting negative results. In investigation planning, site predictability and decision latency mattered more here than the marginal accuracy of the predictor.

10. References

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. https://doi.org/10.1145/2939672.2939785

Ching, J., Uzielli, M., Phoon, K.-K., & Xu, X.-J. (2023). Characterization of autocovariance parameters of detrended cone tip resistance from a global CPT database. *Journal of Geotechnical and Geoenvironmental Engineering*, *149*(10), 04023090. https://doi.org/10.1061/JGGEFK.GTENG-11214

Garnelo, M., Rosenbaum, D., Maddison, C., Ramalho, T., Saxton, D., Shanahan, M., Teh, Y. W., Rezende, D., & Eslami, S. M. A. (2018). Conditional neural processes. *Proceedings of Machine Learning Research*, *80*, 1704–1713. https://proceedings.mlr.press/v80/garnelo18a.html

Guan, Z., Wang, Y., & Zhao, T. (2022). Adaptive sampling strategy for characterizing spatial distribution of soil liquefaction potential using cone penetration test. *Journal of Rock Mechanics and Geotechnical Engineering*, *14*(4), 1221–1231. https://doi.org/10.1016/j.jrmge.2022.01.011

Hollmann, N., Müller, S., Purucker, L., Krishnakumar, A., Körfer, M., Hoo, S. B., Schirrmeister, R. T., & Hutter, F. (2025). Accurate predictions on small data with a tabular foundation model. *Nature*, *637*, 319–326. https://doi.org/10.1038/s41586-024-08328-6

ISSMGE Technical Committee 304. (n.d.). *CPT databases*. Retrieved October 17, 2026, from https://tc304.org/cpt_databases.html

MOS Grondmechanica B.V. (2025, May 1). Offerte A2501409: Realisatie appartementencomplex De Spiegelhoek, Bussum. https://www.spieghelhoek.nl/wp-content/uploads/2025/08/offerte-Mos-Realisatie-appartementencomplex-De-Spiegelhoek-010525-003.pdf

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Prior Labs. (n.d.). *TabPFN* (Version 9.2.0). https://github.com/PriorLabs/TabPFN

Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian processes for machine learning*. MIT Press.

Rijkswaterstaat. (2025, September 1). *New Lock Terneuzen fully operational from 1 August 2025*. https://www.rijkswaterstaat.nl/en/news/archive/2025/09/new-lock-terneuzen-fully-operational-from-1-august-2025

U.S. Government Accountability Office. (2015). *GAO schedule assessment guide: Best practices for project schedules* (GAO-16-89G). U.S. Government Accountability Office. https://www.gao.gov/products/gao-16-89g

U.S. Government Accountability Office. (2020). *Cost estimating and assessment guide: Best practices for developing and managing program costs* (GAO-20-195G). U.S. Government Accountability Office. https://www.gao.gov/products/gao-20-195g